Lesson 3.1

Pairs of Lines & Angles

Parallel, skew, and the angles a transversal makes

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Three ways two lines can relate

Intersecting: the lines meet in one point.

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Parallel: coplanar and **never** meet.

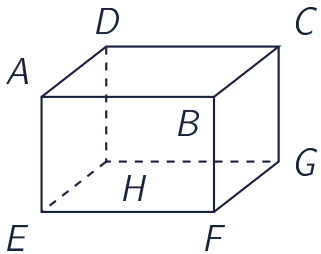
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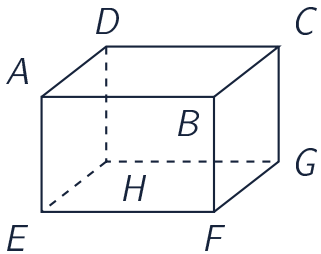
Skew: **not** coplanar and never meet — only in 3D.

Look at a box



$\overleftrightarrow{AB}$ and $\overleftrightarrow{DC}$: parallel

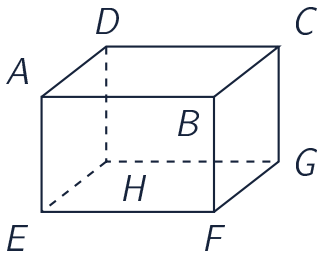
Look at a box



$\overleftrightarrow{AB}$ and $\overleftrightarrow{DC}$: parallel

$\overleftrightarrow{AB}$ and $\overleftrightarrow{BF}$: intersecting

Look at a box

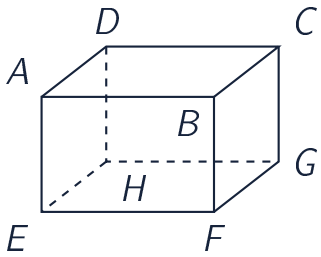


$\overleftrightarrow{AB}$ and $\overleftrightarrow{DC}$: parallel

$\overleftrightarrow{AB}$ and $\overleftrightarrow{BF}$: intersecting

$\overleftrightarrow{AB}$ and $\overleftrightarrow{CG}$: skew

Look at a box



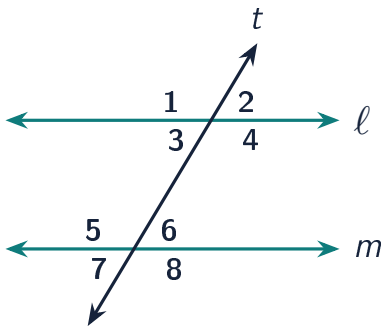
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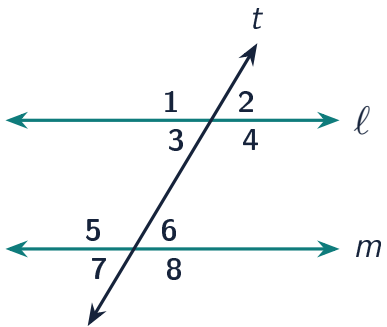
plane $ABCD \parallel$ plane $EFGH$

A transversal makes 8 angles



A **transversal** t crosses two lines and makes **8 angles**.

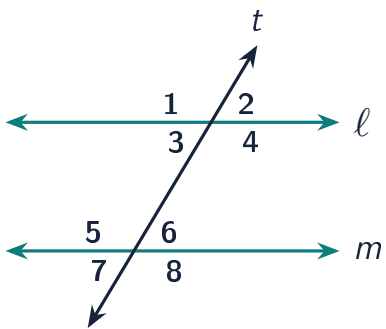
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Angles 1–4 at the top crossing, 5–8 at the bottom crossing.

A transversal makes 8 angles

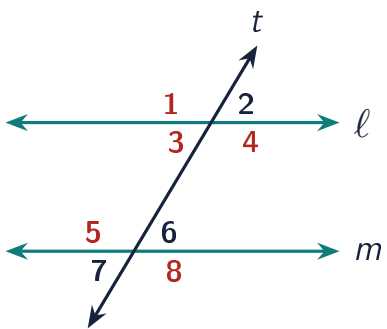


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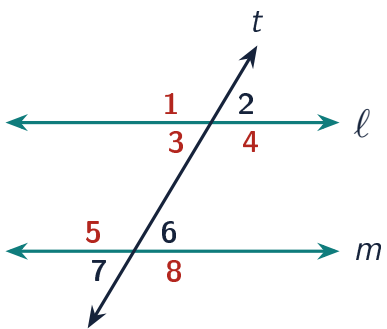
Interior: between the two lines.
Exterior: outside them.

Name the angle pairs



Corresponding: same position, e.g. $\angle 1$ and $\angle 5$

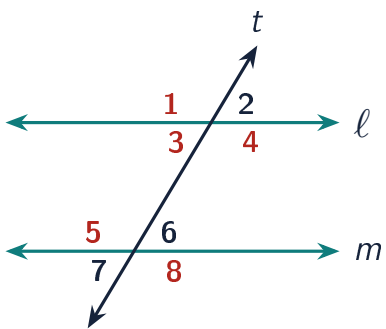
Name the angle pairs



Corresponding: same position, e.g. $\angle 1$ and $\angle 5$

Alternate interior: inside, opposite sides, e.g. $\angle 4$ and $\angle 5$

Name the angle pairs

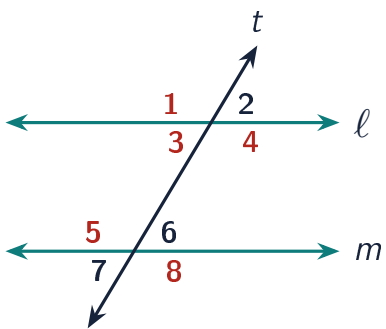


Corresponding: same position, e.g. $\angle 1$ and $\angle 5$

Alternate interior: inside, opposite sides, e.g. $\angle 4$ and $\angle 5$

Alternate exterior: outside, opposite sides, e.g. $\angle 1$ and $\angle 8$

Name the angle pairs



Corresponding: same position, e.g. $\angle 1$ and $\angle 5$

Alternate interior: inside, opposite sides, e.g. $\angle 4$ and $\angle 5$

Alternate exterior: outside, opposite sides, e.g. $\angle 1$ and $\angle 8$

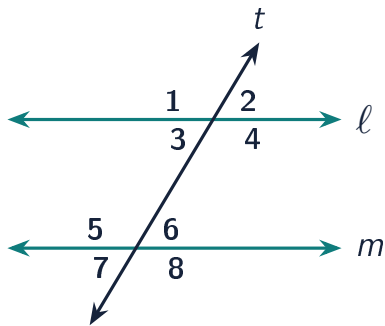
Same-side interior: inside, same side, e.g. $\angle 3$ and $\angle 5$

Watch out

Watch out! Two lines that never meet are not always parallel. If no plane contains both, they are **skew**.

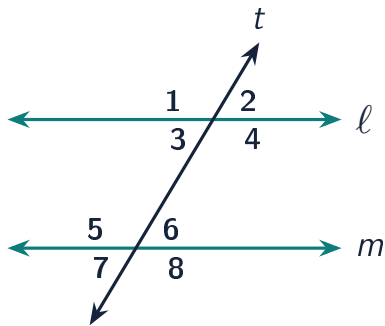
Your turn! Pause the video and try it.

Name each pair: (a) $\angle 2$ and $\angle 6$
(b) $\angle 3$ and $\angle 6$ (c) $\angle 4$ and $\angle 6$



Your turn! Pause the video and try it.

Name each pair: (a) $\angle 2$ and $\angle 6$
(b) $\angle 3$ and $\angle 6$ (c) $\angle 4$ and $\angle 6$



(a) corresponding
(b) alternate interior
(c) same-side interior

Remember

- Lines can be **intersecting**, **parallel** or **skew**.
- A **transversal** makes 8 angles.
- Pairs: **corresponding**, **alternate interior**, **alternate exterior**, **same-side interior**.

Now practice at your level:

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